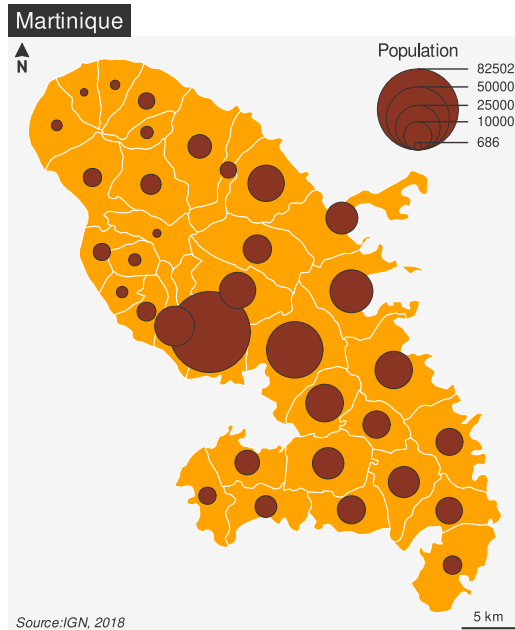


Thematic maps with mapsf :: CHEAT SHEET



Base map

```
library(mapsf)
# import the sample dataset
mtq <- mf_get_mtg()
# display a base map
mf_map(mtg, col = "orange", border = "white")
# display symbology
mf_map(mtg, var = "POP", type = "prop",
       leg_title = "Population", add = TRUE)
# map layout
mf_layout(title = "Martinique",
          credits = "Source:IGN, 2018")
```



Colors

mapsf can use color palettes from `hcl.colors()`
`mf_get_pal()` is useful to create well-balanced
asymmetric diverging palettes

```
mf_get_pal(n = c(7, 2), pal = c("Burg", "Mint"))
```



Symbology

x should be an `sf` object of polygons, lines, or points.

Choropleth (ratios)
`mf_map(x = mtq, var = "var", type = "choro",
 breaks = "quantile")`

Typology (categories)
`mf_map(x = mtq, var = "var", type = "typo")`

Proportional Symbols (stocks)
`mf_map(x = mtq, var = "var", type = "prop",
 inches = .3, symbol = "circle")`

Graduated Symbols (stocks)
`mf_map(x = mtq, var = "var", type = "grad",
 pch = 24)`

Symbols (categories)
`mf_map(x = mtq, var = "var", type = "symb",
 pch = 21:23)`

Choropleth Proportional Symbols (stocks - ratios)
`mf_map(x = mtq, var = c("var1", "var2"),
 type = "prop_choro")`

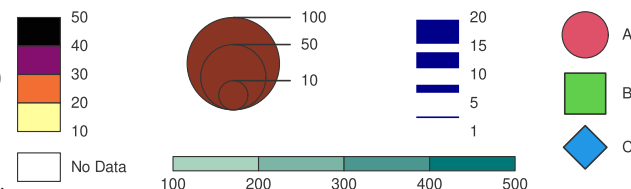
Colorized Proportional Symbols (stocks - categ.)
`mf_map(x = mtq, var = c("var1", "var2"),
 type = "prop_typo")`

Choropleth Symbols (stocks - categ.)
`mf_map(x = mtq, var = c("var1", "var2"),
 type = "symb_choro")`

Raster (single/multi band, continuous, classes, intervals)
`mf_raster(x = raster, type = "continuous")`

Legends

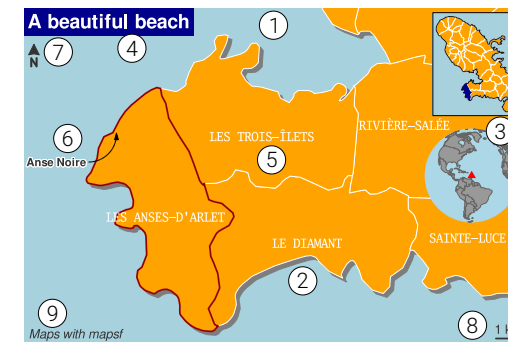
Default legends are plotted with each map.



One can use `mf_legend()` or the `maplegend` package for more customization options.

Map layout

Customization of the layout design



① Set a map theme
`mf_theme(bg = "#aad3df",
 mar = c(0, 0, 0, 0))`

② Create a shadow effect
`mf_shadow( ... )`

③ Create a map inset
`mf_inset_on(mtg)
mf_map(mtg)
mf_inset_off()`
Use `mf_inset_on(x = "worldmap")`
and `mf_worldmap()` to display a globe

④ Display a title
`mf_title( ... )`

More ?

Vignettes on the website:

- > How to use themes
- > How to export maps
- > How to create inset maps
- > How to create faceted maps
- > how to use a custom font family

- ⑤ Display labels
`mf_label( ... )`
- ⑥ Display an annotation
`mf_annotation( ... )`
- ⑦ North arrow
`mf_arrow( ... )`
- ⑧ Scale bar
`mf_scale( ... )`
- ⑨ Credits
`mf_credits( ... )`

Export maps

`mf_export()` exports maps
in PNG and SVG formats.
The width/height ratio of the
exported figure will match
the one of a spatial object.
`mf_export(mtg, width = 3,
 filename = "map.svg")
mf_map(mtg, add = TRUE)
dev.off()`



width : 3 in.
(inferred) height : 3.5 in.

Class intervals

`mf_get_breaks()` uses `classInt`
and additional methods (e.g. "Q6", "geom",
"arith", "msd" or "ckmeans") to classify
continuous variables.

